WORKED SOLUTION

STEP I 2008, Question 10

Mechanics • Projectile Motion

STEP I 2008, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q10

The key idea

Gravity acts only vertically, so the horizontal velocity $V\cos\alpha$ never changes and the range is simply $V\cos\alpha$ times the total time of flight. The time of flight comes from the vertical motion alone, which splits into three stages: up through the bottom layer, the excursion above height h, and back down through the bottom layer. Each stage is a constant-acceleration problem, and symmetry does most of the work.

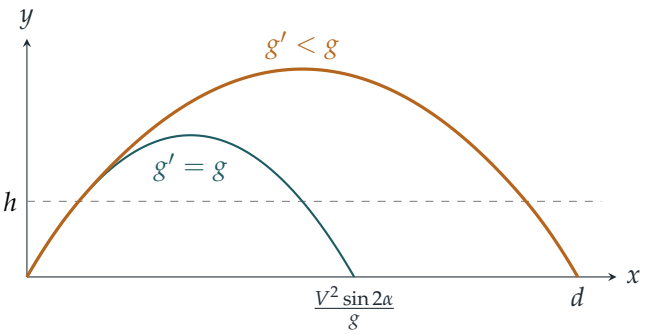
Solution

We model the particle as a point with no air resistance, moving in a vertical plane over flat ground. Take x horizontal and y vertically up, with the origin at the point of projection.

The sketch

Below height h both trajectories obey exactly the same equations, so they coincide there. The condition $V^2\sin^2\alpha > 2gh$ says the initial vertical kinetic energy is enough to rise through height h under acceleration g, so the particle does pass above $y = h$.

Above h, the case $g' < g$ has weaker gravity, so the vertical speed falls off more slowly: the particle climbs higher and spends longer up there, tracing a flatter, wider parabola. The velocity is continuous at $y = h$ (only the acceleration jumps), so the two arcs join smoothly with no corner, although the curvature changes abruptly. On the way down the particle re-enters the bottom layer with the same speed and angle at which it left (by the symmetry of the top arc), and its final arc is a mirror image of the first. So the $g' < g$ path lands further away.



The range

The vertical component of the initial velocity is $V\sin\alpha$, with acceleration $-g$ while $y < h$.

Stage 1: up to height h. Let w be the (upward) vertical speed on reaching $y = h$. By $v^2 = u^2 + 2as$,

$$w^2 = V^2\sin^2\alpha - 2gh = \sin^2\alpha\left(V^2 - 2gh\,\mathrm{cosec}^2\alpha\right) = V'^2\sin^2\alpha,$$

so $w = V'\sin\alpha$, taking the positive root since $\sin\alpha > 0$ and the particle is still rising. (The given condition is exactly what makes V' real and positive.) The time taken is, by $v = u + at$,

$$t_1 = \frac{V\sin\alpha - V'\sin\alpha}{g} = \frac{(V - V')\sin\alpha}{g}.$$

Stage 2: above height h. Now the particle starts at $y = h$ moving upwards at w with acceleration $-g'$. It returns to $y = h$ when $wt - \frac{1}{2}g't^2 = 0$, that is, after

$$t_2 = \frac{2w}{g'} = \frac{2V'\sin\alpha}{g'},$$

and it is then moving downwards with vertical speed w.

Ask yourself Why does the last stage take the same time as the first?

In stage 3 the particle starts at $y = h$ moving downwards at w, with acceleration g downwards, and falls to $y = 0$. That is stage 1 run backwards in time: constant-acceleration motion is reversible, and stage 1 went from $y = 0$ at speed $V\sin\alpha$ to $y = h$ at speed w under the same acceleration. So it reaches the ground with vertical speed $V\sin\alpha$ after time $t_3 = t_1$. (Directly: the vertical speed goes from w to $V\sin\alpha$ at rate g, which takes $(V\sin\alpha - w)/g$.)

The total time of flight is therefore

$$T = 2t_1 + t_2 = 2\sin\alpha\left(\frac{V - V'}{g} + \frac{V'}{g'}\right).$$

There is no horizontal force, so the horizontal velocity stays $V\cos\alpha$ throughout, and

$$\begin{aligned} d = V\cos\alpha \cdot T &= \left(\frac{V-V'}{g} + \frac{V'}{g'}\right) 2V\sin\alpha\cos\alpha \\ &= \left(\frac{V-V'}{g} + \frac{V'}{g'}\right) V\sin 2\alpha, \end{aligned}$$

using $2\sin\alpha\cos\alpha = \sin 2\alpha$. This is the required result.

Checks

- If $g' = g$ the bracket becomes V/g, and $d = V^2\sin 2\alpha/g$, the familiar range on Earth.
- If $g' < g$ then $V'/g' > V'/g$, so d exceeds the uniform-gravity range, matching the sketch.
- If $V^2\sin^2\alpha = 2gh$ exactly then $V' = 0$: the particle just touches $y = h$ and never feels g', and again $d = V^2\sin 2\alpha/g$.
- Numerically, with $V = 10$, $\alpha = 60^\circ$, $g = 10$, $h = 2$ and $g' = 5$ we get $V' = \sqrt{140/3} \approx 6.83$ and $d \approx 14.576$; stepping the motion numerically gives the same landing point to within 10^{-5} relative error. (These are the values used in the sketch.)

Common slips

- Forgetting the descent through the bottom layer, giving $t_1 + t_2$ instead of $2t_1 + t_2$.
- Taking the vertical speed at height h to be V' rather than $V'\sin\alpha$: V' is a speed scaled by $\operatorname{cosec}\alpha$, not the vertical component itself.
- Drawing a corner where the arcs meet: the velocity is continuous, so the join is smooth.

Beyond STEP

Gravity that depends on position is the norm rather than the exception: real gravity weakens as $1/r^2$ with distance from the centre of a planet, and projectile paths become arcs of ellipses rather than parabolas. Splitting a motion into regions with simple force laws and matching position and velocity at the boundaries is a standard technique, used again for refraction of particles and in quantum mechanics with step potentials. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both the constant-acceleration toolkit and orbital motion under inverse-square gravity.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.